Negative Correlation

If two variables tend to move together in opposite direction so that an increase or decrease in the values of one variable is accompanied by a decrease or increase in the value of the other variable, then the correlation is called negative or inverse correlation.

Example

(i) Price and demand

(ii) Repayment period and EMI

(iii) Yield of crops and price

No Correlation

Two variables are said to be uncorrelated if the change in the value of one variable has no connection with the change in the value of the other variable.

For example

We should expect zero correlation (no correlation) between weight of a person and the colour of his hair or the height of a person and the colour of his hair.

Simple correlation

The correlation between two variables is called simple correlation. The correlation in the case of more than two variables is called multiple correlation.

The following are the mathematical methods of correlation coefficient

(i) Scatter diagram

(ii) Karl Pearson's Coefficient of Correlation

9.1.3 Scatter Diagram

Let $(X_1, Y_1), (X_2, Y_2) \ldots (X_N, Y_N)$ be the N pairs of observation of the variables X and Y. If we plot the values of X along x - axis and the corresponding values of Y along y-axis, the diagram so obtained is called a scatter diagram. It gives us an idea of relationship between X and Y. The type of scatter diagram under a simple linear correlation is given below.

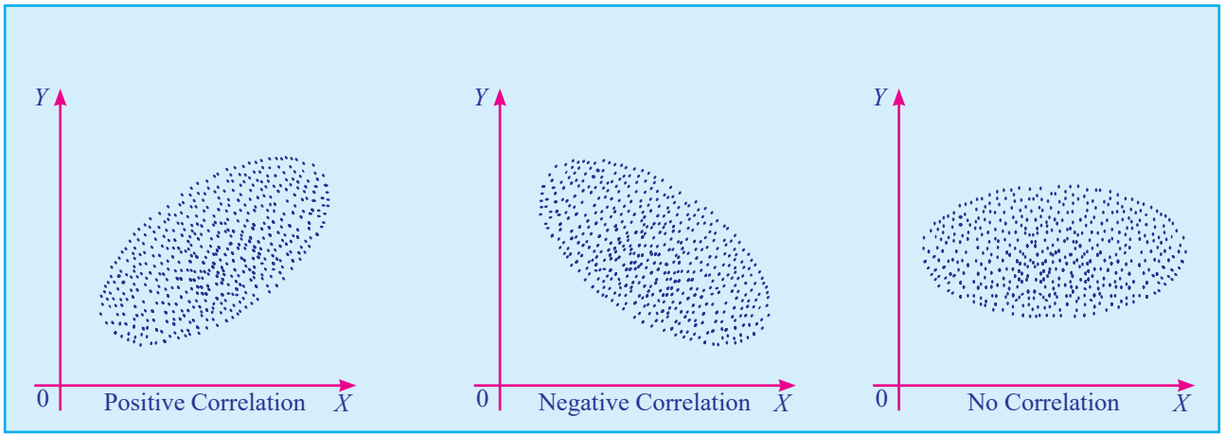


Fig 9.1

(i) If the plotted points show an upward trend, the correlation will be positive.

(ii) If the plotted points show a downward trend, the correlation will be negative.

(iii) If the plotted points show no trend the variables are said to be uncorrelated.

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